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Supplementary Materials for

Tissue-embedded stretchable nanoelectronics reveal endothelial cell–mediated electrical maturation of human 3D cardiac microtissues

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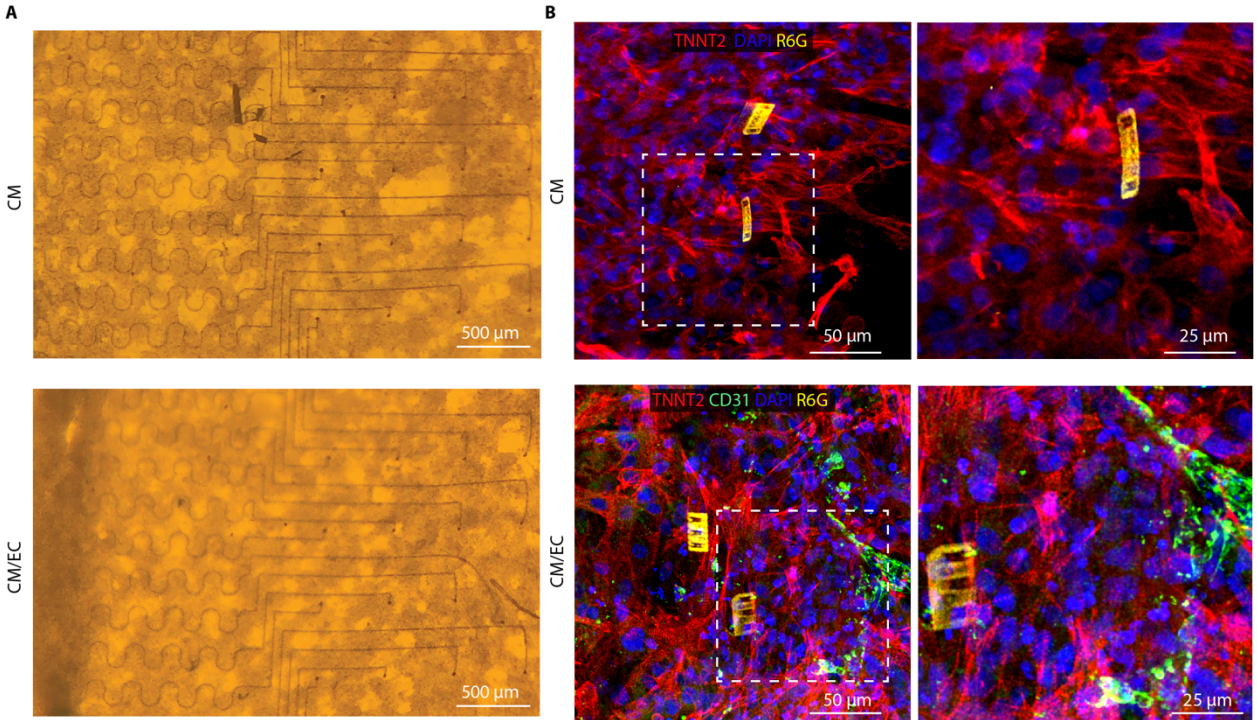


Fig. S1. **Integration of mesh nanoelectronics with hiPSC-CMs or hiPSC-CMs/hiPSC-ECs.** (**A**) Bright field images showing the hiPSC-CMs-electronics hybrid (top) and hiPSC-CM/hiPSC-EC-electronics hybrid (bottom). (**B**) Representative confocal fluorescence images of cleared, immunostained hiPSC-CMs (top) and hiPSC-CM/hiPSC-EC (bottom) electronic hybrids at day 37 of differentiation. Red colors correspond to TNNT2; Green colors correspond to CD31; Blue colors correspond to DAPI; Yellow colors correspond to R6G (E-barcodes of the nanoelectronics labeled by R6G).

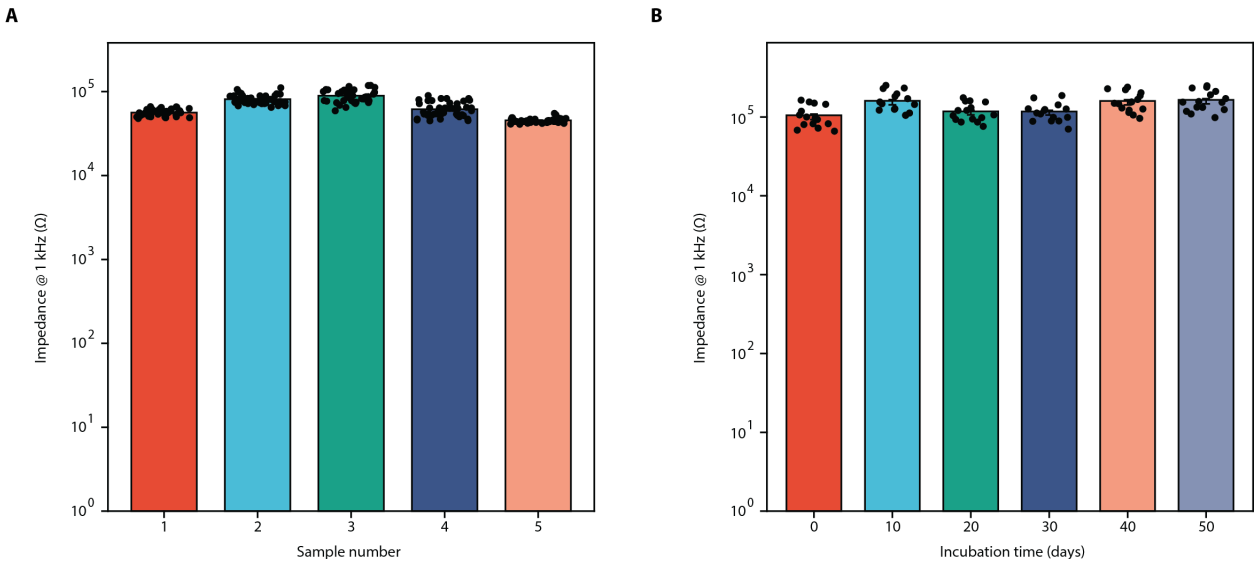


Fig. S2. Chronic electrical performance of stretchable mesh nanoelectronics in physiological condition.

(**A**) Statistics of electrochemical impedance of electrodes at 1 kHz across five representative mesh nanoelectronics and (**B**) longevity test of the electrochemical impedance of electrodes at 1 kHz for 50 days of incubation in PBS at 37 °C from 15 representative samples (values are mean ± s.e.m.).

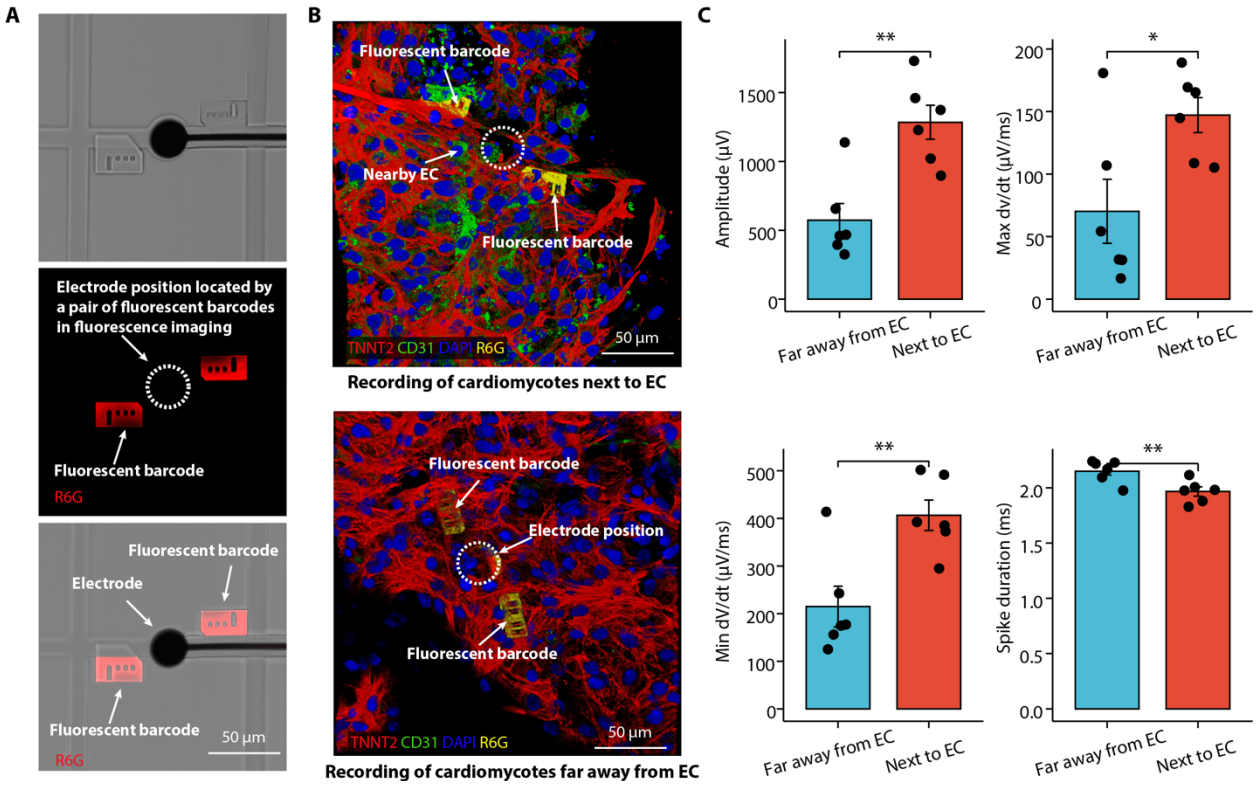


Fig. S3. hiPSC-ECs improved electrical maturation of the nearby hiPSC-CMs
(**A**) Representative bright field (top), fluorescence (middle), and overlay (bottom) images showing the relations of paired fluorescent barcode and corresponding electrode location. (**B**) Representative 3D reconstructed fluorescence images in the hiPSC-CM/hiPSC-EC samples showing the recording of cardiomyocytes next to EC (top) and far away from EC (bottom). Dash circle boxes highlight the electrode positions. (**C**) Comparison of spike waveform features of CMs far from ECs and CMs next to ECs in the hiPSC-CM/hiPSC-EC sample. Data are mean ± s.e.m., *p<0.05, **p<0.01 by two-tailed, unpaired t-test.

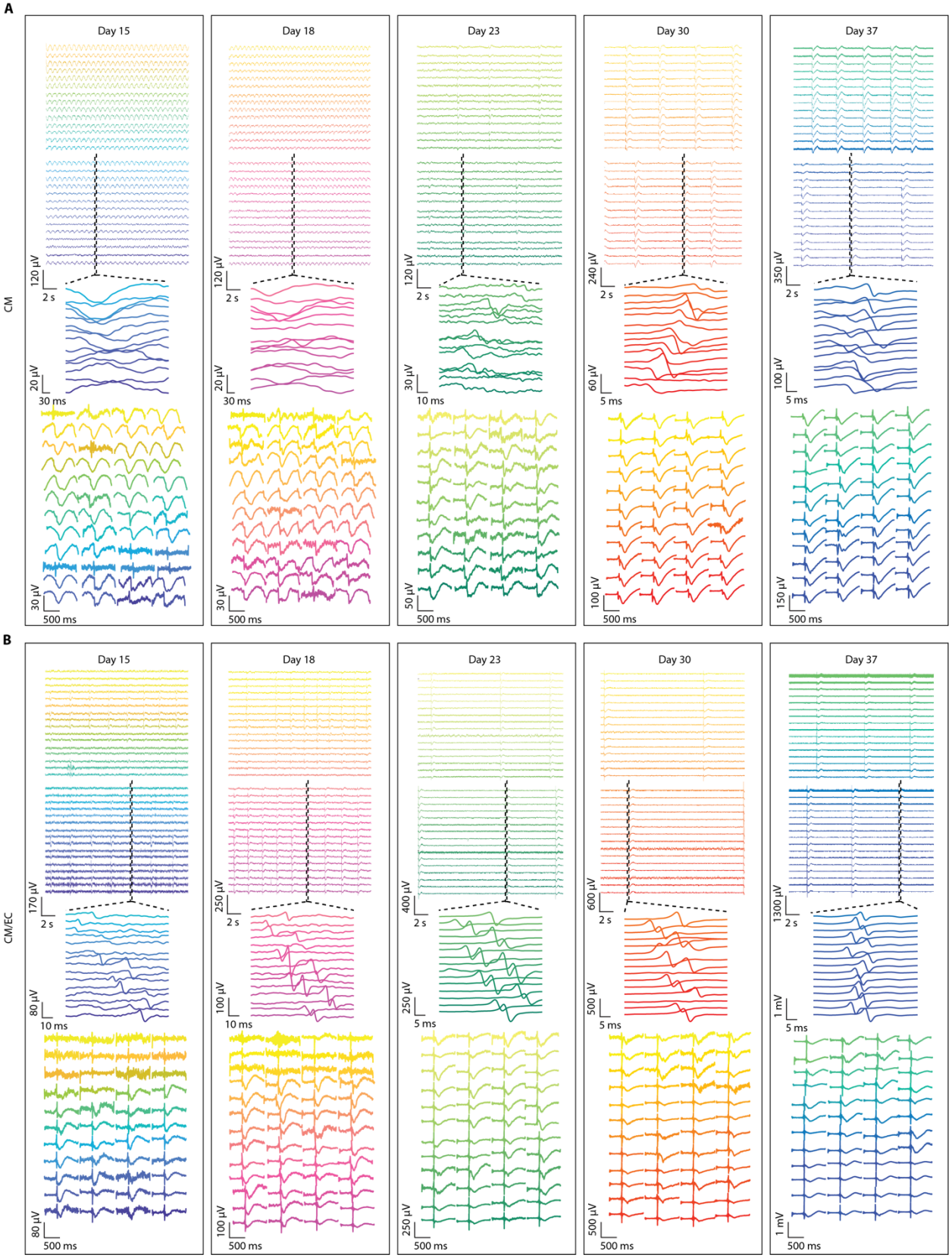

A
CM
Day 15
Day 18
Day 23
Day 30
Day 37
120 µV
2 s
20 µV
30 ms
30 µV
500 ms
120 µV
2 s
20 µV
30 ms
30 µV
500 ms
120 µV
2 s
30 µV
10 ms
50 µV
500 ms
240 µV
2 s
60 µV
5 ms
100 µV
500 ms
350 µV
2 s
100 µV
5 ms
150 µV
500 ms
B
CM/EC
Day 15
Day 18
Day 23
Day 30
Day 37
170 µV
2 s
80 µV
10 ms
80 µV
500 ms
250 µV
2 s
100 µV
10 ms
100 µV
500 ms
400 µV
2 s
250 µV
5 ms
250 µV
500 ms
600 µV
2 s
500 µV
5 ms
500 µV
500 ms
1300 µV
2 s
1 mV
5 ms
1 mV
500 ms

Fig. S4. Long-term, continuous electrical recording of hiPSC-CMs and hiPSC-CMs/hiPSC-ECs at different differentiation times.
(**A-B**) Representative raw voltage traces (top), zoom-in traces highlighted by dashed boxes (middle), and extracted channel-averaged spike waveforms (bottom) showing the distinct electrical activities at day 15, day 18, day 23, day 30, and day 37 of differentiation for hiPSC-CMs (**A**) and hiPSC-CMs/hiPSC-ECs (**B**).

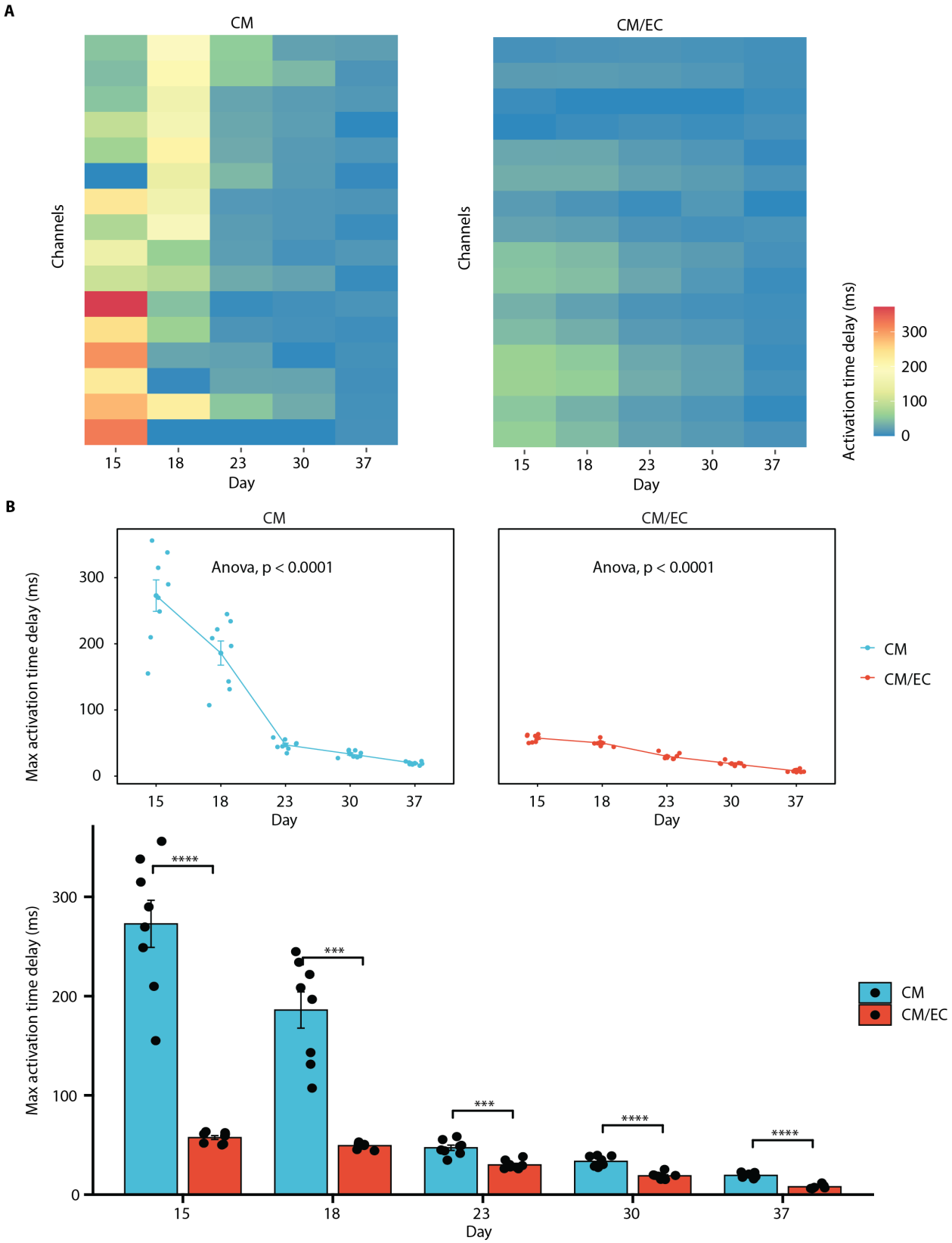

A
CM
CM/EC
Channels
Channels
15
18
23
30
37
Day
Day
Activation time delay (ms)
300
200
100
0
B
CM
CM/EC
Anova, p < 0.0001
Anova, p < 0.0001
Max activation time delay (ms)
300
200
100
0
15
18
23
30
37
Day
Day
CM
CM/EC
****
***
***
****
****
Max activation time delay (ms)
300
200
100
0
15
18
23
30
37
Day
CM
CM/EC

Fig. S5. Activation time delay of hiPSC-CMs and hiPSC-CMs/hiPSC-ECs at different differentiation times.

(**A**) Representative heatmaps showing the activation delay across different channels at different differentiation times of hiPSC-CMs and hiPSC-CMs/hiPSC-ECs. (**B**) Line plots (top) and bar plots (bottom) showing the maximum activation time delay across channels at different differentiation times of hiPSC-CMs and hiPSC-CMs/hiPSC-ECs. Data are mean ± s.e.m., ***$p<0.001$, ****$p<0.0001$ by two-tailed, unpaired t-test for barplots; One-way anova test for line plots.

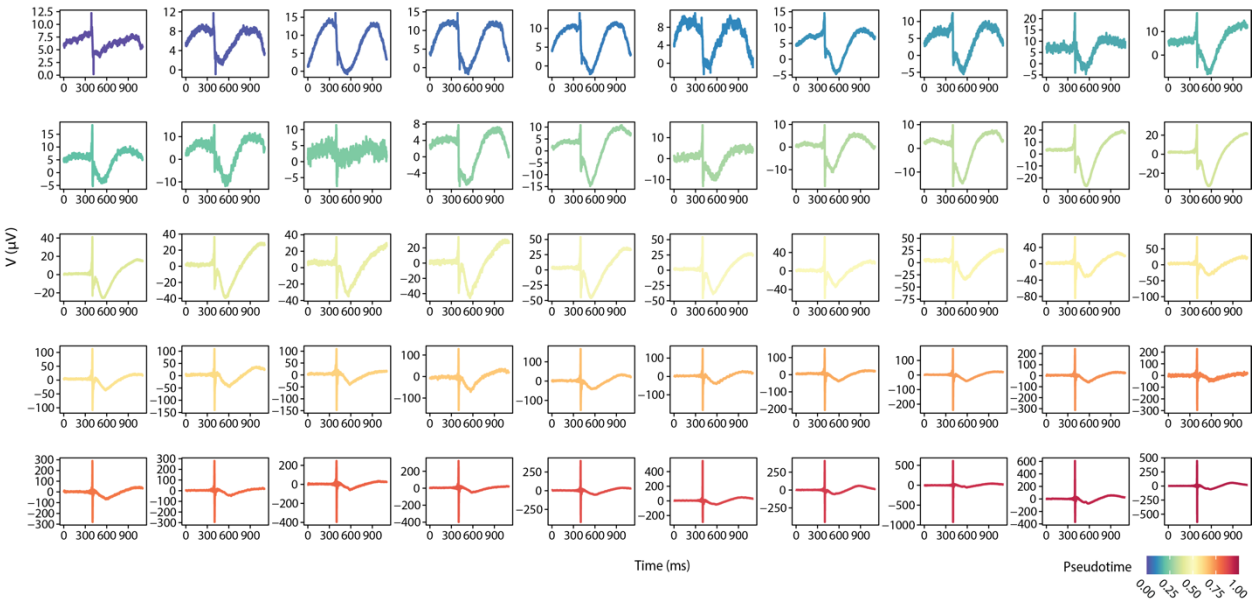


Fig. S6. **Evolution of extracellular field potential spike waveforms ordered by pseudotime**. Spike waveforms ordered and colored by the pseudotime bin. The pseudotime of all spike waveforms is divided into 50 bins and the averaged spike waveform of the spikes within the bin is shown for the corresponding bin.

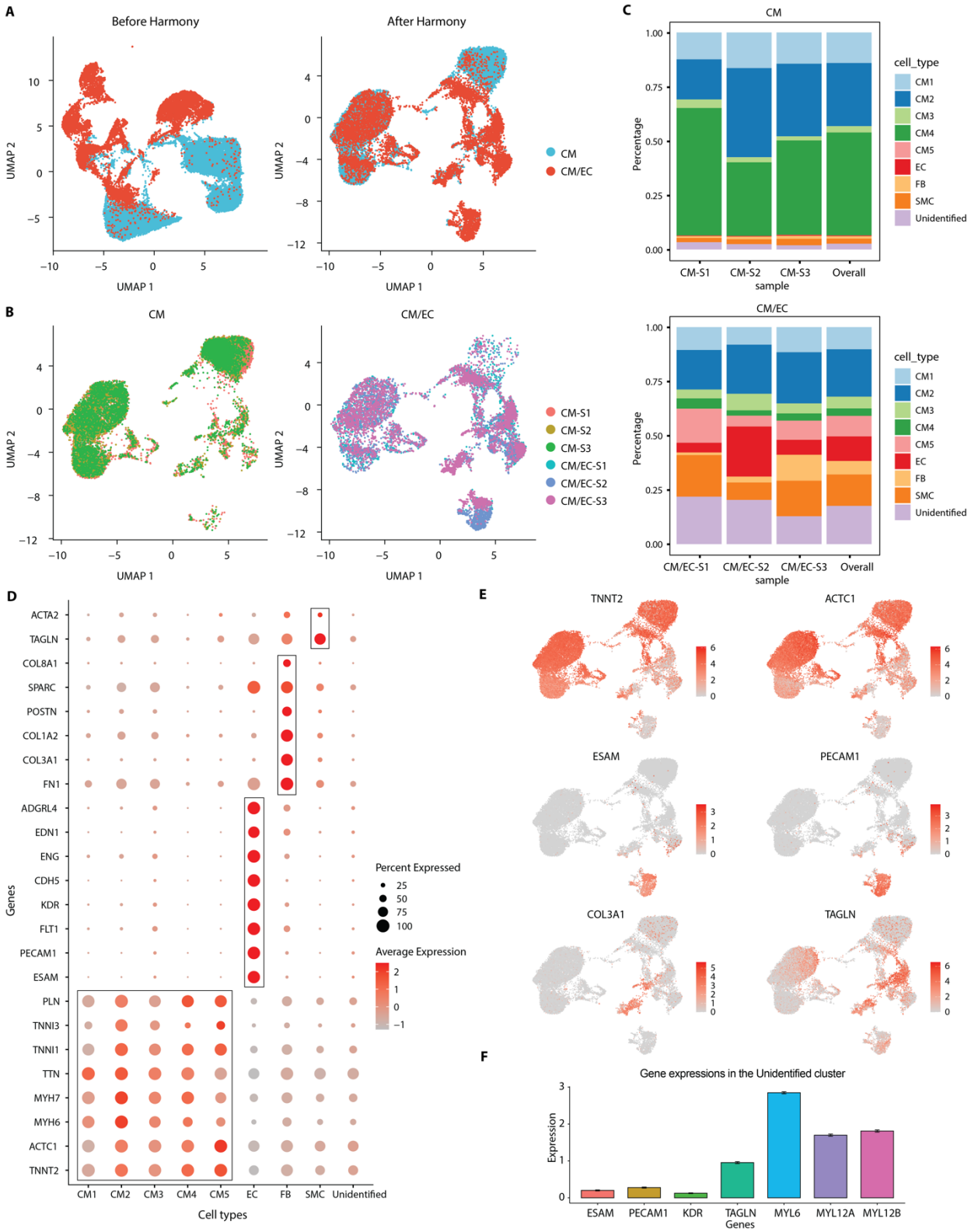

A
Before Harmony
After Harmony
UMAP 1
UMAP 2
CM
CM/EC
B
CM
CM/EC
CM-S1
CM-S2
CM-S3
CM/EC-S1
CM/EC-S2
CM/EC-S3
C
CM
CM/EC
Percentage
sample
cell_type
CM1
CM2
CM3
CM4
CM5
EC
FB
SMC
Unidentified
CM-S1
CM-S2
CM-S3
Overall
CM/EC-S1
CM/EC-S2
CM/EC-S3
D
Genes
Cell types
Percent Expressed
Average Expression
ACTA2
TAGLN
COL8A1
SPARC
POSTN
COL1A2
COL3A1
FN1
ADGRL4
EDN1
ENG
CDH5
KDR
FLT1
PECAM1
ESAM
PLN
TNNI3
TNNI1
TTN
MYH7
MYH6
ACTC1
TNNT2
E
TNNT2
ACTC1
ESAM
PECAM1
COL3A1
TAGLN
F
Gene expressions in the Unidentified cluster
Expression
ESAM
PECAM1
KDR
TAGLN
MYL6
MYL12A
MYL12B
Genes

Fig. S7. Data integration and gene expression analysis of scRNA-seq data.
(**A**) UMAP visualization of gene expression before (left) and after (right) Harmony data integration of hiPSC-CMs and hiPSC-CMs/hiPSC-ECs. Cells are colored by datasets. (**B**) UMAP visualization of gene expression after data integration for three hiPSC-CM samples (left) and three hiPSC-CM/hiPSC-EC samples (right). Cells are colored by samples. (**C**) Cell type percentages in hiPSC-CMs (top) or hiPSC-CMs/hiPSC-ECs (bottom) across samples. (**D**) Dot plot showing the marker gene expressions for each cell type. (**E**) UMAP plots showing the selected marker gene expression. (**F**) Barplot showing the gene expression in the Unidentified cluster.

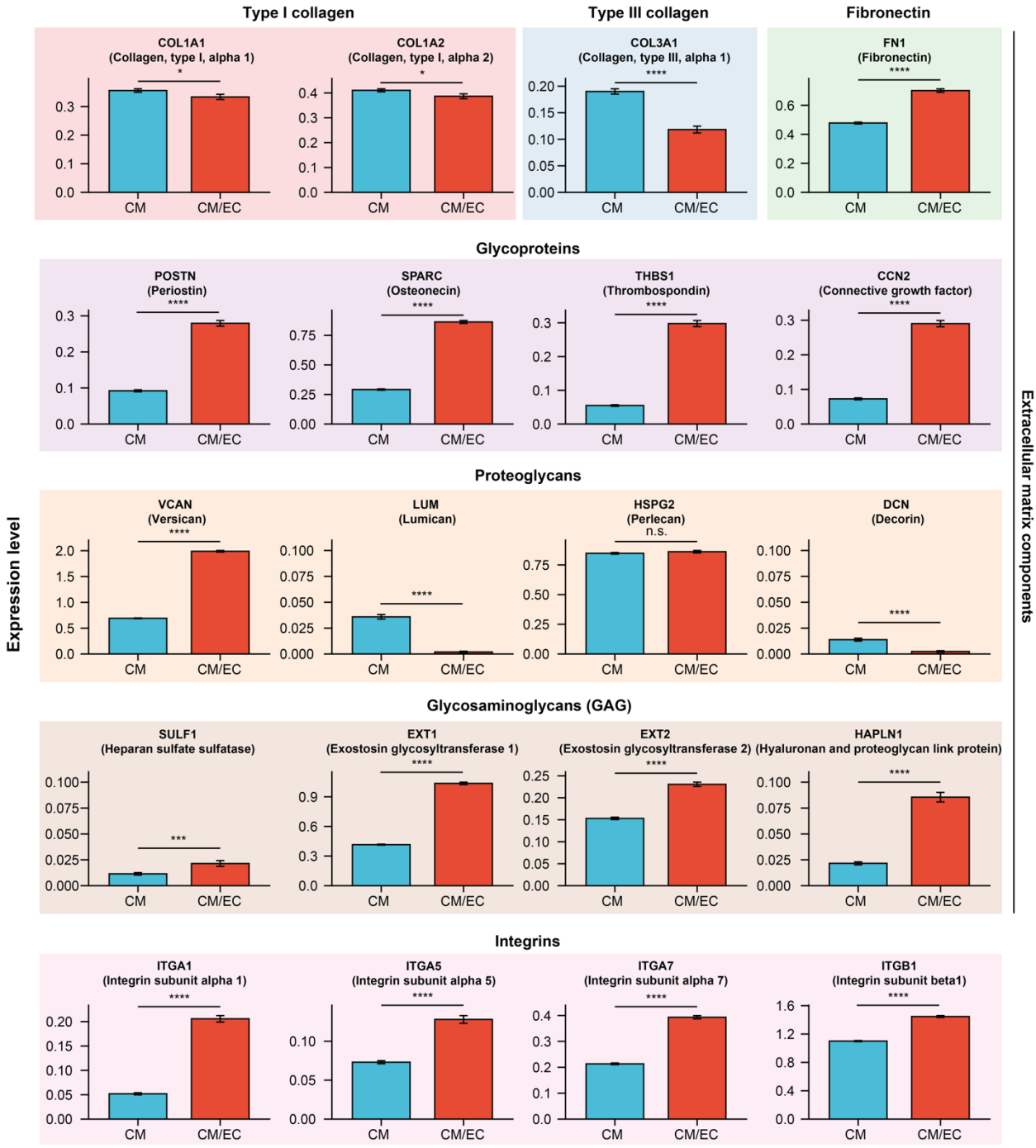


Fig. S8. HiPSC-ECs changed the expressions of extracellular matrix in hiPSC-CMs.
Barplots showing selected extracellular matrix and integrin gene expressions in cardiomyocytes, comparing hiPSC-CMs and hiPSC-CMs/hiPSC-ECs. Extracellular matrix genes were grouped according to different extracellular matrix components, including collagen, fibronectin, glycoprotein, proteoglycan, and glycosaminoglycans. Data are mean ± s.e.m., *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

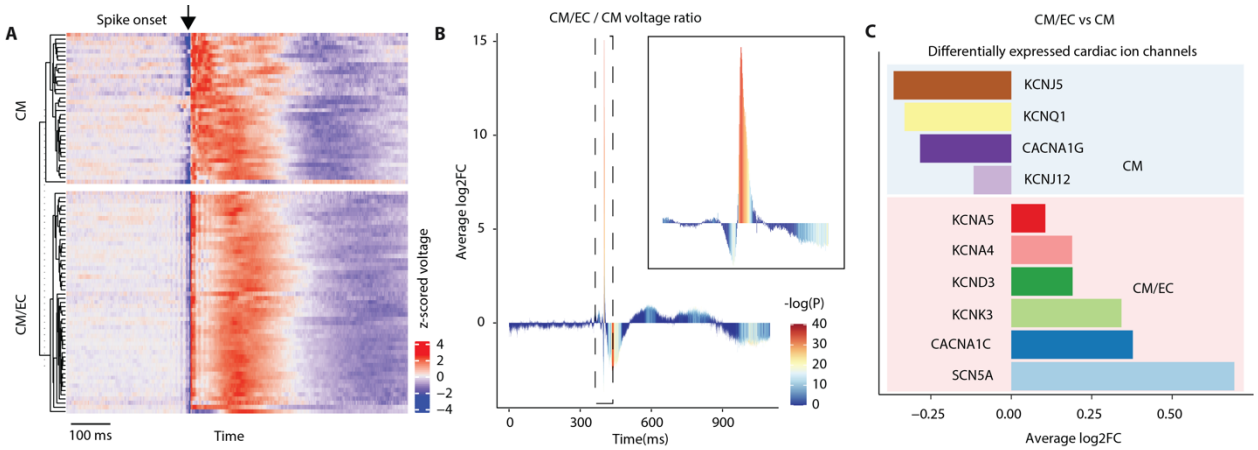


Fig. S9. HiPSC-ECs changed the expressions of ion channels in hiPSC-CMs.
(**A**) Heatmap showing the row z-scored voltage for hiPSC-CMs and hiPSC-CMs/hiPSC-ECs for samples at day 23 of differentiation. (**B**) Barplot showing the average log fold change of the voltage as a function of time by comparing hiPSC-CMs/hiPSC-ECs to hiPSC-CMs. Inset shows the zoom-in of the time window around spike initiation. (**C**) Barplot showing the differentially expressed cardiac ion channels in CM by comparing hiPSC-CMs/hiPSC-ECs to hiPSC-CM.

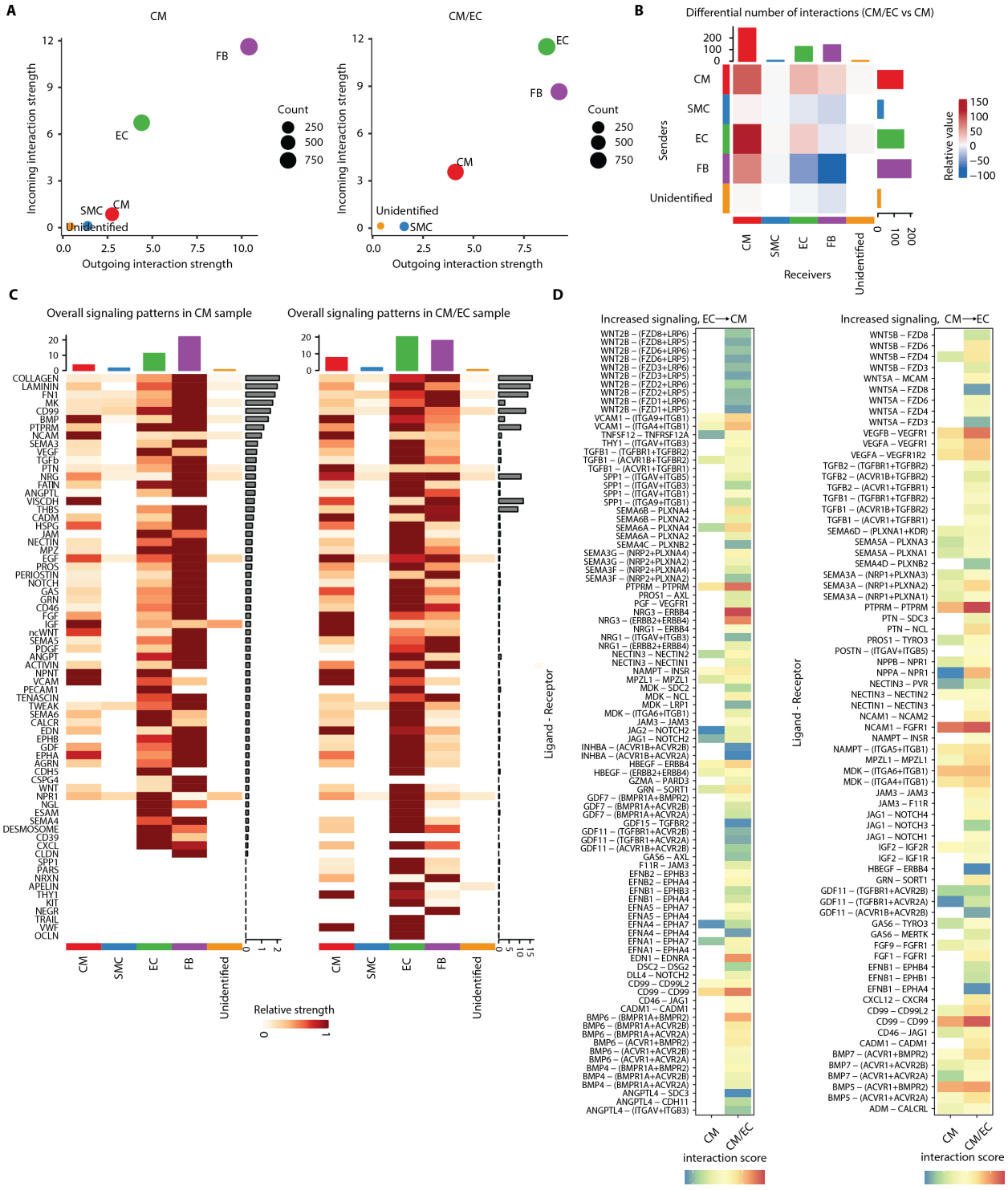

A
CM
CM/EC
Incoming interaction strength
Outgoing interaction strength
Count
250
500
750
FB
EC
SMC
CM
Unidentified
B
Differential number of interactions (CM/EC vs CM)
Senders
Receivers
Relative value
C
Overall signaling patterns in CM sample
Overall signaling patterns in CM/EC sample
Ligand - Receptor
Relative strength
D
Increased signaling, EC⟶CM
Increased signaling, CM⟶EC
interaction score

Fig. S10. Cell-cell interaction analysis by CellChat.
(**A**) Scatter plots showing the incoming and outgoing interaction strength for different cell types in hiPSC-CMs (left) and hiPSC-CMs/hiPSC-ECs (right). (**B**) Differential number of interactions between different cell types comparing hiPSC-CMs/hiPSC-ECs to hiPSC-CMs. (**C**) Heatmap showing the overall signaling patterns for hiPSC-CMs (left) and hiPSC-CMs/hiPSC-ECs (right). (**D**) Heatmap showing the upregulated ligand-receptor pairs from EC to CM (left) and from CM to EC (right) by comparing hiPSC-CMs/hiPSC-ECs to hiPSC-CMs.

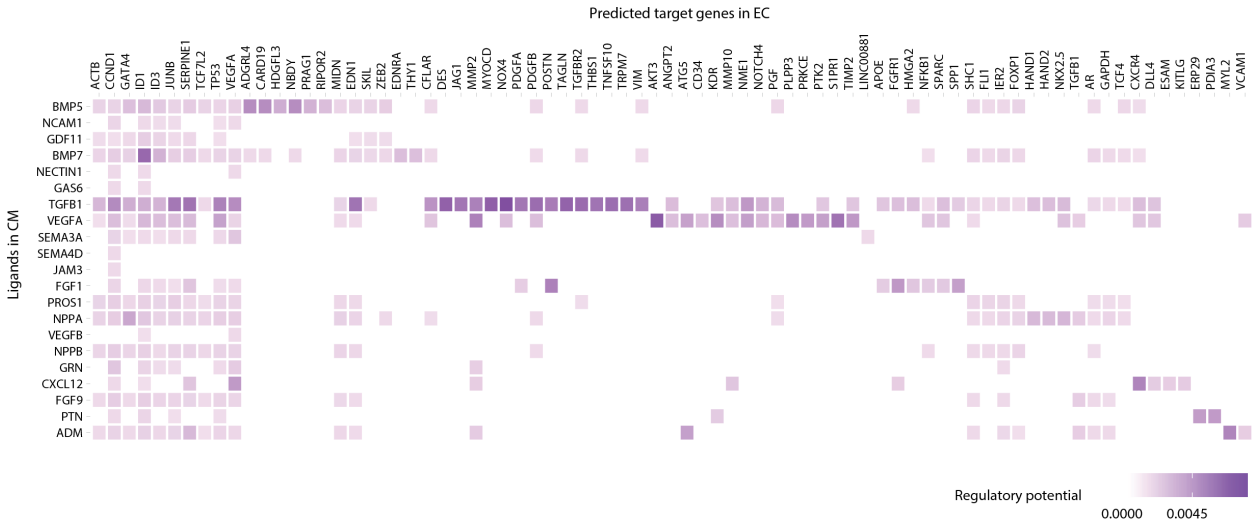


Fig. S11. Ligand-target analysis by NicheNet using CM as sender and EC as receiver. Ligand-target matrix showing the regulatory potential between sender (CM) ligand and predicted receiver (EC) target genes analyzed by NicheNet.

Table S1. Comparisons of different methods for recording the hPSC-CM electrophysiology.

	Device category	Recording longevity	Sensor contacts with cells	Max time delay* (ms)	Estimated conduction velocity* (m/s)	Sampling rate (kHz)	Stretchability	Electrical property	Ref.
Planar MEA	2D rigid	~ 1 week	2D surface	5-10	0.03-0.39	NA	Not stretchable	NA	42, 43
3D-SR-BAs	3D flexible	~ 3 hours	3D surface	~1	~0.12	20	Not stretchable	Impedance 10-100 kΩ @ 1 kHz	31
Nanowire FET arrays	3D flexible	~ 1 week	3D multiple layers	~30	~0.28	20-100	Not stretchable	Transconductance 10-25 μS/V	38
MicroECP	2D flexible	NA	2D tissue patch	~10	~0.17	10	Not stretchable	Impedence ~0.1 kΩ @ 1 kHz	21
Ultrasoft electronics	2D flexible	~ 4 days	2D surface	22-30	0.15-0.3	NA	~ 35.9% max tensile strain	Impedence~3 kΩ @ 1 kHz	39, 40
Nanotransistor sensor	3D rigid	NA	3D bottom surface	~15	~0.04	30	Not stretchable	Transconductance ~4.2 μS/V	41
Stretchable nanoelectronics	3D flexible	~ month	3D tissue-wide	10-270	0.01-0.23	30	~50% compressive strain ~20% tensile strain	Impedance 10-100 kΩ @ 1 kHz	This work

* The maximum time delay and conduction velocity of hPSC-CM vary depending on their differentiation stage, with early immature hPSC-CM showing a high maximum time delay and low conduction velocity. Our study tracked the electrical signals of hPSC-CM over time and found that the maximum time delay gradually decreases as the hPSC-CM mature. Other studies typically measure hPSC-CM at later stages with lower maximum time delay.